

Noise-Robust Spoken Digit Recognition Using PCEN–MFCC Feature Fusion

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Abstract

This work studies noise-robust spoken digit recognition under a fixed-model constraint: we improve the input representation while keeping the baseline bidirectional LSTM classifier unchanged. Our front end first trims low-energy leading and trailing regions, then extracts two complementary feature streams from each utterance. The first stream uses per-channel energy normalization (PCEN), which adaptively suppresses slowly varying background energy. The second stream uses mel-frequency cepstral coefficients (MFCCs), which compactly represent the speech spectral envelope. For both streams, we append delta and delta-delta coefficients and concatenate them into a 231-dimensional frame-level feature vector. Trained on clean speech and evaluated on clean as well as babble-noise-corrupted speech at 10 dB and 5 dB SNR, the proposed system achieves 98.67% accuracy on clean data, 91.67% at 10 dB, and 81.00% at 5 dB. These results show that careful feature design can substantially improve robustness to noise even without changing the downstream sequence model.

Index Terms: Spoken digit recognition, noise robustness, PCEN, MFCC, BiLSTM

1. Introduction

Robust speech recognition remains difficult when training and test conditions do not match. Even for a small-vocabulary task such as isolated spoken digit recognition, background noise can significantly degrade performance by corrupting the acoustic cues used for classification. This makes spoken digit recognition a useful controlled setting for studying how front-end representations behave under noise [1].

In this project, the mismatch is explicit: the classifier is trained on clean speech and evaluated on both clean utterances and babble-noise-corrupted utterances at 10 dB and 5 dB SNR. Under these conditions, a standard MFCC-based front end provides a strong clean-speech baseline because MFCCs offer a compact representation of the short-time speech spectrum and spectral envelope [2], but such representations become much less reliable as noise increases in mismatched conditions [3]. Since the assignment constrains the classifier to remain fixed, the main design question is whether a more robust acoustic representation can recover performance without changing the downstream sequence model.

To address this, we replace the MFCC-only input with a fused feature representation that combines MFCC and per-channel energy normalization (PCEN) features together with temporal derivatives. MFCCs provide a compact description of the speech spectral envelope, while PCEN offers adaptive normalization that is better suited to noisy conditions and was introduced as a dynamic alternative to static log-mel compression

[4, 5]. The goal of the paper is to evaluate whether these complementary representations improve robustness under clean-to-noisy train–test mismatch.

The rest of the paper is organized as follows. Section 2 reviews prior work on spoken digit recognition and noise-robust front-end methods. Section 3 describes the proposed system, experimental setup, results, and runtime. Section 4 summarizes the main findings, discusses their implications, and outlines limitations and future directions.

2. Background and related work

Isolated and connected digit recognition have long been used as controlled benchmarks for automatic speech recognition because they preserve many of the core challenges of ASR including speaker variability, temporal dynamics, and acoustic mismatch while keeping the output vocabulary small. Much of the classical literature on noisy digit recognition was organized around the TIDIGITS/AURORA framework, where standardized noisy digit corpora and fixed HMM back ends allowed researchers to compare front-end robustness methods directly [1, 6, 3]. Within this setting, mel-frequency cepstral coefficients (MFCCs) became a standard baseline because they provide a compact representation of the short-time speech spectrum and the vocal-tract spectral envelope [2]. However, work on AURORA-style noisy digit tasks also showed that conventional cepstral features degrade sharply under additive noise and channel mismatch [6, 3].

A large body of research therefore focused on improving the acoustic front end rather than changing the recognizer itself. Early examples include RASTA processing, which suppresses slowly varying spectral components and channel effects [7]. Later work introduced more explicitly noise-robust auditory-inspired features such as power-normalized cepstral coefficients (PNCC), which combine asymmetric temporal filtering, power-law compression, and temporal masking to improve robustness relative to MFCC- and PLP-style front ends [8]. Beyond single-stream cepstral features, other studies explored richer spectro-temporal or exemplar-based representations. For example, exemplar-based sparse methods were shown to improve automatic speech recognition under additive noise by modeling noisy speech as a combination of speech and noise exemplars [9]. A consistent message across this literature is that front-end design remains critical when test conditions differ from training conditions.

More recently, open spoken-digit corpora have made small-vocabulary speech recognition easier to study in low-resource settings. The Free Spoken Digit Dataset (FSDD), used in this project, provides an English 8 kHz corpus of trimmed spoken digits and has become a common benchmark for rapid prototyp-

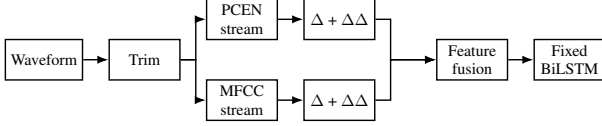


Figure 1: *Proposed front-end pipeline. The classifier is kept fixed; only the feature extraction stage is modified.*

ing of spoken-digit classifiers [10]. In parallel, modern spoken-digit systems have increasingly adopted neural back ends. Examples include deep-learning approaches for spoken English digits [11], MFCC-based CNN models for Pashto isolated digits [12], and CNN-based Amharic spoken-digit recognition [13]. These studies show that modern classifiers can perform well even on relatively small digit corpora, but they also reinforce that performance remains strongly dependent on the front-end representation supplied to the model.

The present work is most closely related to research that improves robustness through the feature pipeline. In particular, per-channel energy normalization (PCEN) was introduced as a dynamic alternative to static log-mel compression for robust and far-field keyword spotting [4]. Subsequent analysis explained why PCEN can be advantageous in noisy acoustic pattern-recognition tasks: it performs temporal smoothing, adaptive gain control, and nonlinear compression independently in each channel, thereby suppressing slowly varying background energy more effectively than fixed log compression [5]. Unlike work that changes the classifier architecture, applies multi-condition training, or relies on extensive augmentation, this report keeps the baseline BiLSTM and clean-only training protocol fixed. Our goal is therefore narrower and more controlled: to test whether combining MFCC-based features with a parallel PCEN stream and temporal derivatives can improve spoken digit recognition under clean-to-noisy train-test mismatch.

3. Project description

3.1. Feature extraction

The baseline system uses 13-dimensional MFCC features extracted at a 100 Hz frame rate and feeds them to a bidirectional LSTM classifier. In this work, we keep the classifier fixed and redesign only the acoustic front end in order to improve robustness under noisy test conditions.

Figure 1 summarizes the proposed pipeline. Each waveform is first converted to floating-point format and trimmed using `librosa.effects.trim` with `top_db=40`. This removes low-energy leading and trailing regions, reducing non-informative padding and variability at utterance boundaries.

Two complementary feature streams are then extracted from the trimmed signal. The first stream is based on per-channel energy normalization (PCEN) [4, 5]. We compute a 64-channel mel spectrogram using a 25 ms window, a 10 ms hop, and a 256-point FFT, then apply PCEN with a time constant of 0.4. The resulting 64-dimensional PCEN representation is mean-normalized, and first- and second-order temporal derivatives are appended, giving

$$64 + 64 + 64 = 192 \quad (1)$$

features per frame.

The second stream is based on mel-frequency cepstral coefficients (MFCCs). Using the same short-time analysis front end, we compute 13 MFCC coefficients and apply cepstral mean

and variance normalization (CMVN). First- and second-order derivatives are then appended, producing

$$13 + 13 + 13 = 39 \quad (2)$$

features per frame.

Using the same window length, hop size, and FFT size in both branches ensures that the PCEN and MFCC streams are aligned in time and can be concatenated directly at the frame level. The derivatives are computed separately within each branch before fusion, so the final representation preserves both the base features and their local temporal changes.

The final frame-level representation is obtained by concatenating the two streams:

$$192 + 39 = 231. \quad (3)$$

This fused representation provides the classifier with both a compact cepstral description of the spectral envelope and a more noise-tolerant energy-normalized representation.

3.2. Classifier and implementation

The fused feature sequence is passed to the provided bidirectional LSTM classifier. The model contains two bidirectional LSTM layers with hidden size 128 and dropout 0.2. The recurrent output is aggregated using mean pooling over valid, non-padded time steps only, after which a small classification head consisting of `Linear(256, 32)`, `ReLU`, dropout 0.3, and `Linear(32, 10)` produces the final 10-class prediction.

The system is implemented in Python using PyTorch, torchaudio, and librosa. Training is performed in Google Colab with GPU acceleration. The main hyperparameters are batch size 32, 40 epochs, Adam optimization with learning rate 3×10^{-4} , and gradient clipping with max norm 1.0. A fixed random seed is used to improve reproducibility. To ensure a fair comparison with the baseline and satisfy the project constraints, we keep the BiLSTM architecture, learning rate, batch size, and clean-only training setup unchanged, and modify only the feature extraction stage.

The experiments use the Free Spoken Digit Dataset (FSDD) [10], with 2,500 clean training utterances and 300 utterances in each test condition. The model is trained only on clean speech and evaluated on clean test speech as well as babble-noise-corrupted test speech at 10 dB and 5 dB SNR. The task is utterance-level 10-class classification. The final fused representation has dimension 231 per frame, with mean sequence lengths of 43.4 frames for the training set, 42.8 for the clean test set, and 43.6 for both noisy test sets.

3.3. Results and runtime

Table 1 compares the baseline system with the proposed fused-feature system. The proposed front end improves performance in all conditions, with the largest gains appearing under noisy evaluation. The baseline accuracies reported in Table 1 correspond to the provided MFCC + BiLSTM baseline under the same evaluation conditions.

Relative to the baseline, the proposed system improves clean accuracy by 1.00 percentage point, 10 dB accuracy by 22.00 percentage points, and 5 dB accuracy by 30.00 percentage points. These gains indicate that the fused PCEN-MFCC representation is substantially more robust to acoustic mismatch than MFCCs alone.

Figure 2 shows the confusion matrices for the final model. Performance remains close to perfect in the clean condition and

Table 1: Classification accuracy (%) for the baseline and proposed systems.

Method	Clean	10 dB	5 dB
Baseline MFCC + BiLSTM	97.00	70.00	51.00
Proposed fused features + BiLSTM	98.67	91.67	81.00

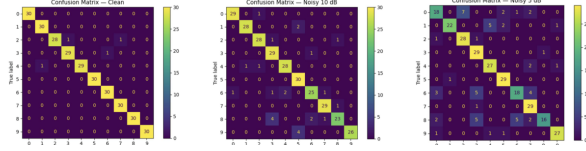


Figure 2: Confusion matrices for the final model on clean, 10 dB, and 5 dB test sets.

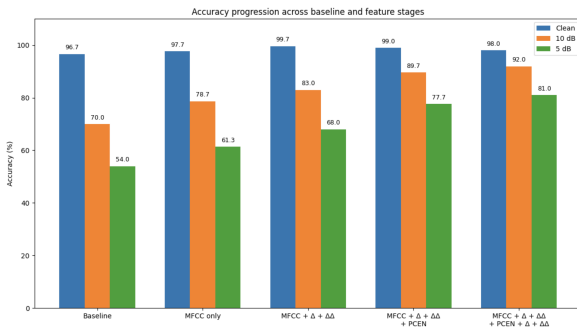


Figure 3: Accuracy progression across the baseline and successive feature refinements.

Table 2: Runtime of the proposed system on Google Colab.

Stage	Time (s)
Feature loading/extraction	122.9
Training	85.6
Evaluation	0.3
Total	208.8

strongly diagonal at 10 dB. At 5 dB, the number of off-diagonal errors increases, but the system remains substantially more stable than the MFCC-only baseline.

Figure 3 shows the progression across intermediate feature configurations. Adding temporal derivatives improves robustness relative to static features, and adding the PCEN stream provides a further substantial gain in the noisy conditions. The final fused representation gives the best overall balance between clean and noisy performance.

Table 2 reports the runtime of the complete pipeline on Google Colab, including feature loading and extraction, training, and final evaluation. Feature extraction accounts for most of the execution time, while evaluation itself is comparatively inexpensive once the model is trained.

4. Summary and discussion

The results of this project show that the acoustic front end plays a major role in robustness when training and test conditions are mismatched. The proposed fused PCEN-MFCC rep-

resentation improves performance over the MFCC-only baseline in all three evaluation conditions, but the size of the gain is not uniform. The clean-speech accuracy increases only modestly, from 97.00% to 98.67%, whereas the gains under noise are much larger: the 10 dB accuracy increases from 70.00% to 91.67%, and the 5 dB accuracy increases from 51.00% to 81.00%. This uneven improvement is important because it suggests that the proposed features are not simply making the representation larger or more expressive in a generic sense. Instead, they are addressing the specific weakness of the baseline under noisy test conditions. In other words, the main benefit of the proposed front end is robustness to acoustic mismatch rather than an across-the-board increase in model capacity.

The improvement pattern is also consistent with the structure of the feature pipeline. The MFCC stream provides a compact summary of the short-time spectral envelope, which remains a useful descriptor of the phonetic content of isolated spoken digits. However, MFCCs are derived from mel-filterbank energies that are easily distorted by additive noise, especially when training is performed only on clean speech. The PCEN stream helps compensate for this weakness by introducing temporal smoothing, adaptive gain control, and nonlinear compression independently in each mel channel. As a result, slowly varying background energy is suppressed more aggressively than in a standard log-compressed front end. This is especially relevant for babble noise, which can mask low-energy speech components and short transitions. By combining MFCC and PCEN features, the classifier receives both a compact spectral description and a representation that is more stable under background interference.

The role of temporal derivatives is also supported by the progression across intermediate systems, but their effect depends on which feature stream they are applied to. In the MFCC branch, adding delta and delta-delta coefficients improves performance, indicating that short-time dynamics are already important for the task. This makes sense for spoken digits, since many distinctions rely not only on steady-state vowels but also on brief consonantal events and transitions. Digits such as “three,” “five,” and “nine” contain relatively weak or short-duration cues that can be obscured under babble noise, so including temporal derivatives helps preserve information about how the spectral content evolves frame to frame. However, the ablation results also show that once the PCEN stream is introduced, clean-speech performance decreases slightly: clean accuracy drops from 99.7% for MFCC + Δ + $\Delta\Delta$ to 99.0% after adding PCEN, and then to 98.0% in the final fused system with PCEN + Δ + $\Delta\Delta$. A plausible explanation is that PCEN is designed primarily to improve robustness in noisy or far-field conditions rather than to optimize already-clean speech [4, 5]. Because PCEN applies adaptive gain control, temporal smoothing, and nonlinear compression, it can reshape fine spectral detail that is already informative in clean utterances. Adding delta and delta-delta features on top of the PCEN stream may further emphasize small frame-to-frame fluctuations introduced by this adaptive normalization. In other words, temporal derivatives appear beneficial for the MFCC stream, but when applied to the PCEN branch they introduce a small clean-speech tradeoff while still contributing to much larger gains under noise. The final fused representation therefore benefits from both dynamic information and channel-wise robustness, even though that benefit is expressed much more strongly in noisy conditions than in clean speech.

The confusion matrices provide a complementary view of the results across noise conditions. In the clean condition, clas-

sification is nearly perfect and the matrix is strongly diagonal, which indicates that the fixed BiLSTM is already adequate when the acoustic representation is clean. At 10 dB, the matrix remains mostly diagonal, suggesting that the proposed front end preserves most class-discriminative structure even when moderate babble noise is present. At 5 dB, the number of off-diagonal errors increases substantially, which is expected because low-SNR babble can mask short consonantal events and reduce spectral contrast. Some of the remaining confusions at 5 dB are also phonetically plausible. In particular, digits whose identity depends strongly on fricative onset cues, such as zero, six, seven, and three, may be especially vulnerable once babble obscures the onset spectrum: fricatives are characterized by broad-band noise with important high-frequency structure, and the acoustic contrast among fricatives depends in part on differences in that spectral shape. This helps make sense of broader spreading for digits like six, which is confused with zero, three, and seven at 5 dB, because once the onset is partially masked the classifier must rely more heavily on the remaining vowel and coda information. Vowel identity may also contribute to several of the off-diagonal patterns, since the first two formants encode major aspects of vowel height and backness; when babble smears F1/F2 structure and formant transitions, words that are distinguished mainly by their vowel nucleus or vowel-to-consonant trajectory can become harder to separate. That may help explain why eight spreads across several other classes at 5 dB, and why some confusions do not cluster around a single competing digit. These pairwise explanations should be treated as plausible phonetic interpretations of the matrix rather than as direct proof, but overall the confusion patterns still suggest that the fused representation preserves useful structure under severe degradation instead of simply overfitting to the clean training condition.

Another useful takeaway is that these gains are achieved without modifying the network architecture, training hyperparameters, or clean-only training protocol. This makes the comparison to the baseline especially meaningful. The improvement cannot be attributed to a deeper or larger model, more aggressive optimization, or exposure to noisy examples during training. Instead, the results isolate the contribution of feature design itself. For a small-vocabulary speech task, this is a valuable result: it shows that careful handcrafted front-end processing can still provide substantial benefits even when the back end is a modern recurrent neural model. At the same time, the runtime measurements show that this improvement is not free. Most of the total execution time is spent in feature loading and extraction rather than in model evaluation, reflecting the added computational cost of the richer front end. Even so, the total runtime remains modest in the context of a Colab-based experiment, which suggests that the robustness gains are achieved at a reasonable computational cost for a course-scale system.

Despite these strengths, the study has several limitations. First, the task is restricted to isolated spoken digits from a relatively small corpus, so the findings may not transfer directly to larger-vocabulary or continuous-speech recognition problems. Second, the model is trained only on clean speech, which is useful for exposing robustness weaknesses but does not show whether the same feature design would remain optimal under multi-condition training or data augmentation. Third, model selection is based on noisy-condition development performance rather than on a dedicated validation split, which weakens the strictness of the evaluation protocol. These limitations do not invalidate the observed gains, but they do mean that the conclusions should be interpreted as strong evidence within the

scope of the project rather than as a final statement about general speech-recognition performance.

Several directions for future work follow naturally from these observations. A more rigorous next step would be to introduce a separate validation split and repeat the experiments over multiple random seeds in order to measure the stability of the gains. It would also be useful to perform a more systematic ablation study with matched training conditions, comparing MFCC-only, PCEN-only, MFCC+derivatives, and fused PCEN–MFCC variants to isolate the contribution of each component more precisely. Another important extension would be to combine the current front end with noise augmentation or multi-condition training, which would help determine whether feature-level robustness and training-time robustness provide complementary benefits. Finally, the same front-end design could be evaluated on larger and more diverse datasets to test whether the improvement observed here extends beyond isolated digit recognition. Overall, the project supports a clear conclusion: even in a neural speech-classification pipeline, front-end feature engineering remains a powerful tool for improving robustness when the acoustic environment at test time differs from that seen during training.

5. References

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